WINDOWS 11

Windows 11 System Administration & Network Security Suite

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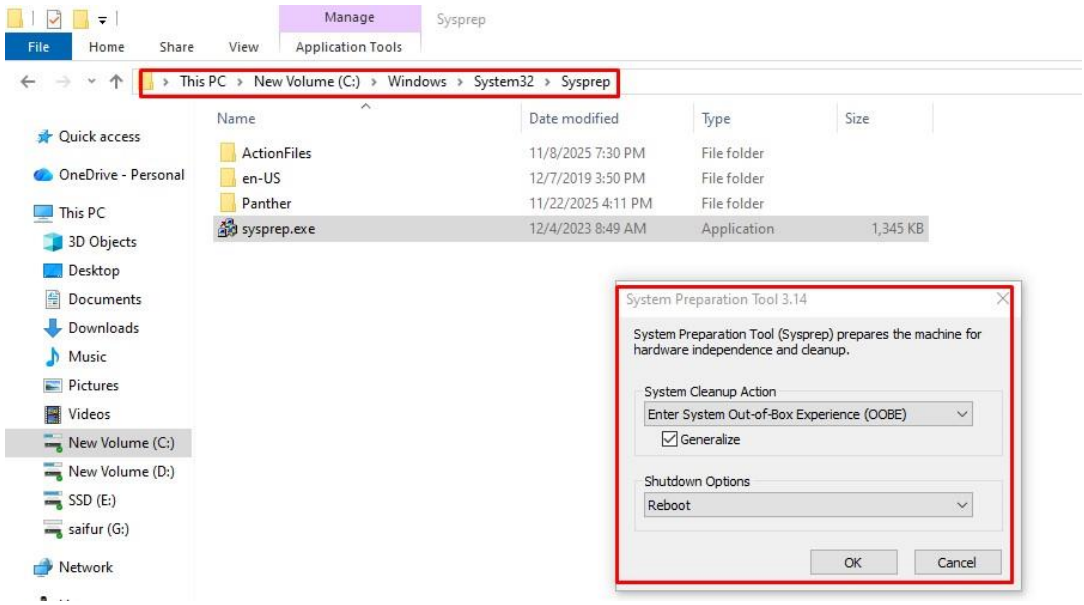
Network and System Admin

Task 1: Simplified Windows Deployment Documentation

This method involves creating a **"Master Image"** and deploying it to the remaining PCs using DISM commands for 50 users.

Step 1: Prepare the Master PC (Sysprep)

After installing and configuring the necessary software on a reference PC, it must be generalized for deployment. Because Sysprep Remove hardware-specific information (like the SID).



The function of the parameter generalize is to prepare the image for deployment on any hardware. The function of the parameter shutdown is to shut down and restart the PC after the process is complete.

Capture the Image (DISM Capture)

The shut-down reference PC must be booted from **Windows PE (Preinstallation Environment)** to create a WIM file of its Windows partition. For, Save the contents of the Windows drive into a single .wim file.

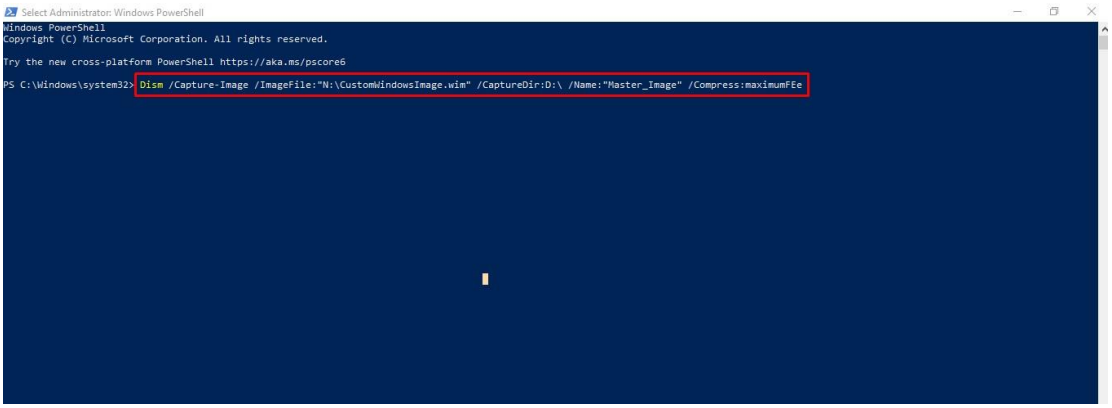
Command (in WinPE):

Dism /Capture-Image /ImageFile:N:\CustomWindowsImage.wim" /CaptureDir:D:\ /Name:"Master_Image" /Compress:maximum

Here is a short key-point explanation:

1. dism /capture-image → capture a partition into a WIM file
2. /imagefile:N:\CustomWindowsImage.wim → save the captured image to N: as MasterImage.wim
3. /capturedir:D: → capture everything from drive D:

4. /name:"MasterImage" → name the image “MasterImage” inside the WIM file 5. /Compress:maximum → Applies maximum compression to reduce the WIM file size.



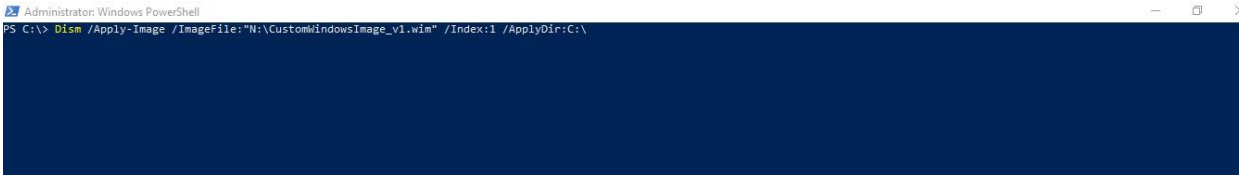
Parameter ImageFile Function is The name and network path of the WIM file.
Parameter ImageFile Function is The drive from which the Windows image is being captured.

Step 3: Deployment (DISM Apply)
Each target PC must be booted from **Windows PE**, and the WIM file must be installed using the *DISM /Apply-Image* command.

Prepare Disk (Diskpart)

Prepare the hard disk of each new PC to be bootable:
Diskpart>select disk 0 >clean >(create partitions) exit

Apply Image Install the captured WIM file onto the target PC:



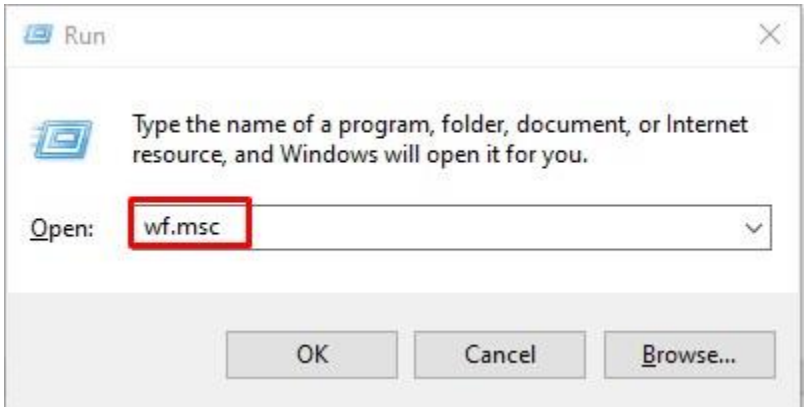
1. /imagefile:N:\CustomWindowsImage.wim → the WIM file you want to apply.
2. /index:1 → choose image number 1 inside the WIM file.
3. /applydir: C: → apply (install/restore) the image onto drive “C”

Create Boot Files Make
Windows bootable:
W:\Windows\System32\bcdboot W:\Windows /s S: /f UEFI
After deployment, when the PC reboots, each user will encounter the OOBE (Out-of-Box Experience) screen and can set up their own user account.

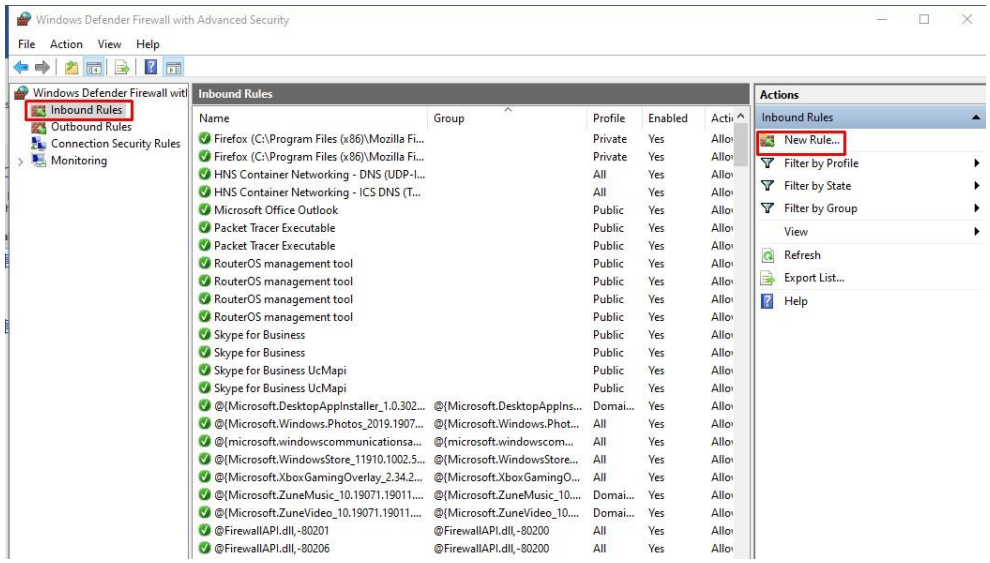
Task 2: Isolating a Specific system on the LAN (Windows Firewall):

This process uses the Windows Defender Firewall with Advanced Security to block incoming common services like Ping, File Sharing, and Remote Desktop Protocol (RDP), while allowing the system to maintain outbound internet access.
Blocking Incoming Ping (ICMP) from the LAN . To specifically block ping requests from other systems on your local network, follow these steps:

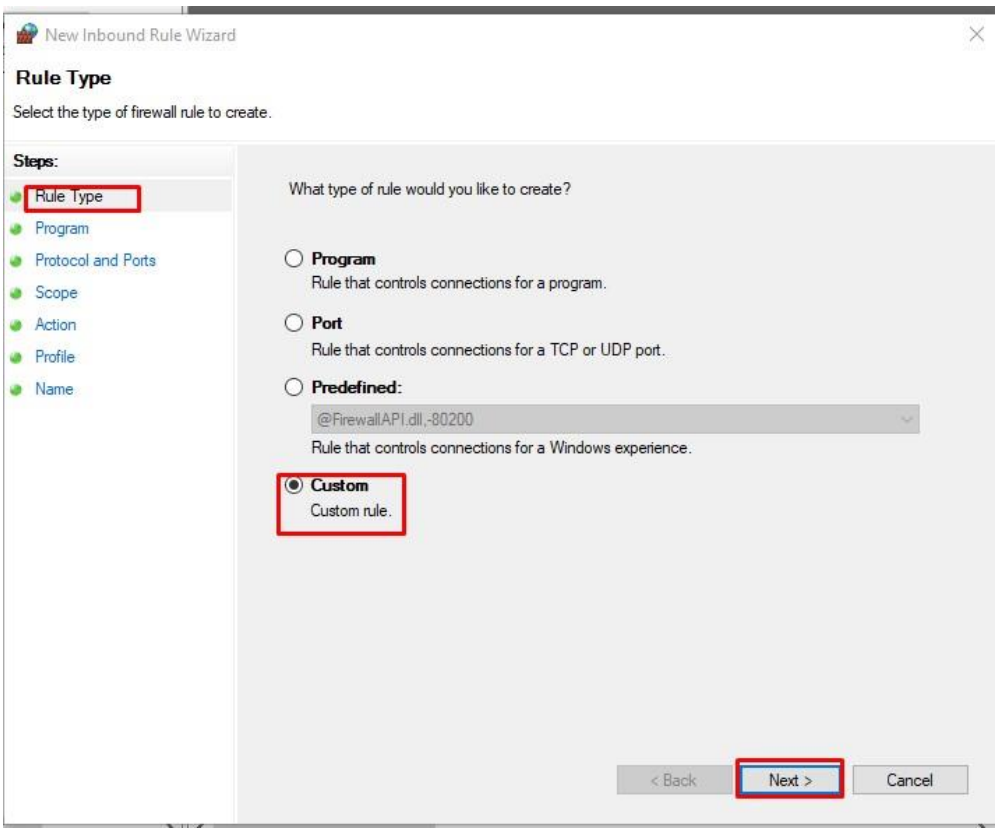
Press **Win + R**, type **wf.msc**, and press **Enter**. Open Windows Defender Firewall with Advanced Security.



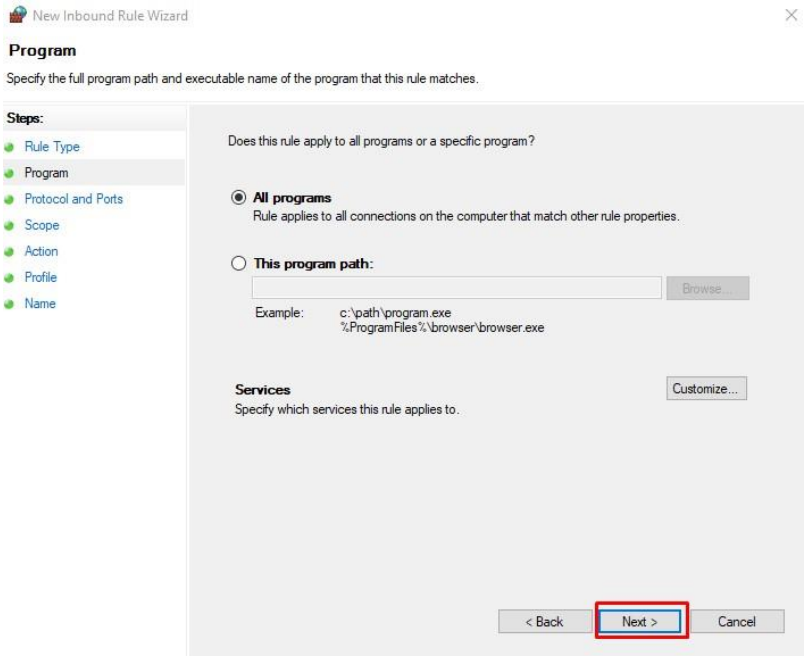
go to the right-hand "**Actions**" panel and click on "**New Rule**".



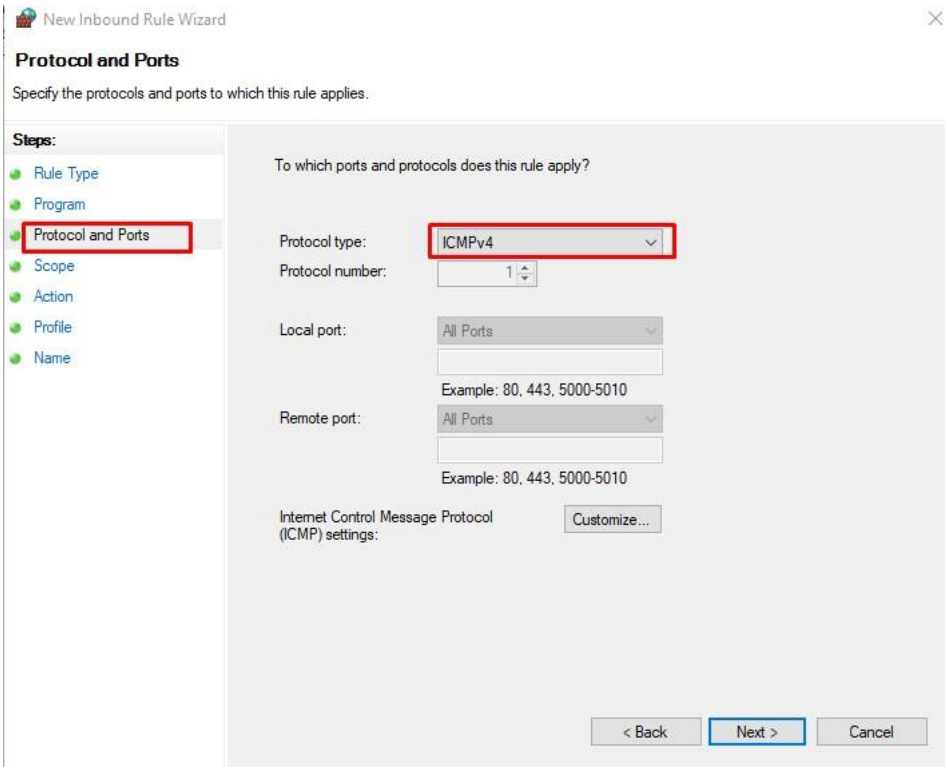
For Rule Type, select "**Custom**" and click Next.



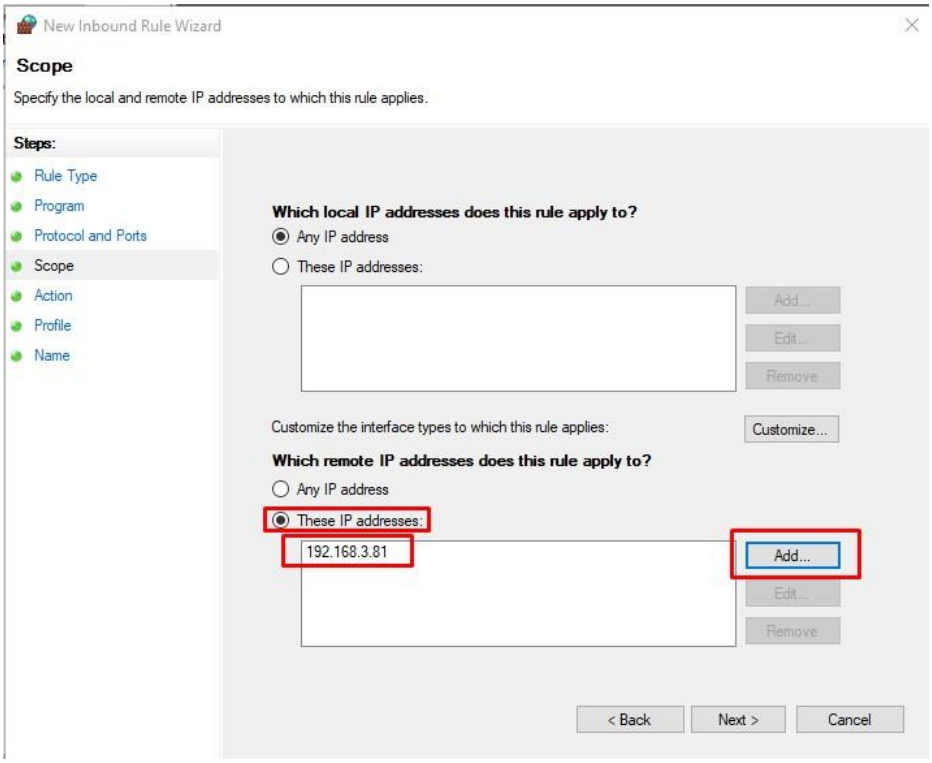
On the Program and Protocol and Ports screens, do not make any changes. Simply click Next on each.



In the Protocol and Ports section: Set the Protocol type: to "**ICMPv4**".

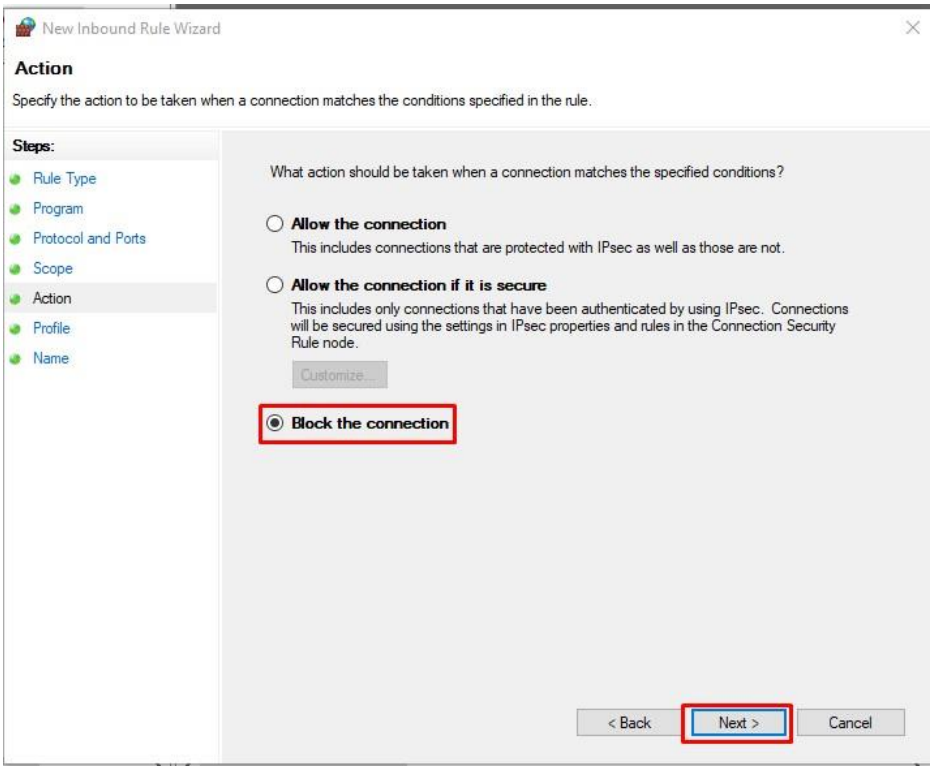


Select the option **"These IP addresses".**Click the "Add..." button and enter LAN's network subnet (e.g., 192.168.3.81). then Click **Add** and **Next**.

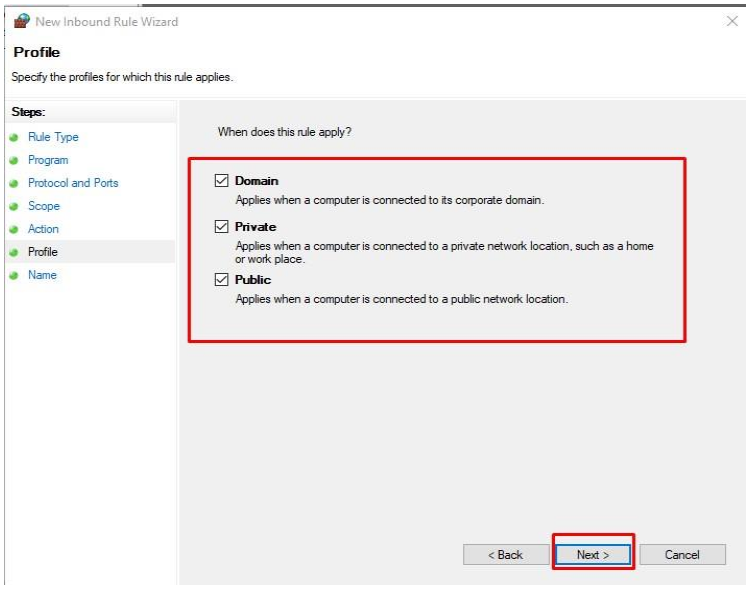


This ensures the rule only blocks ping requests originating from within LAN.

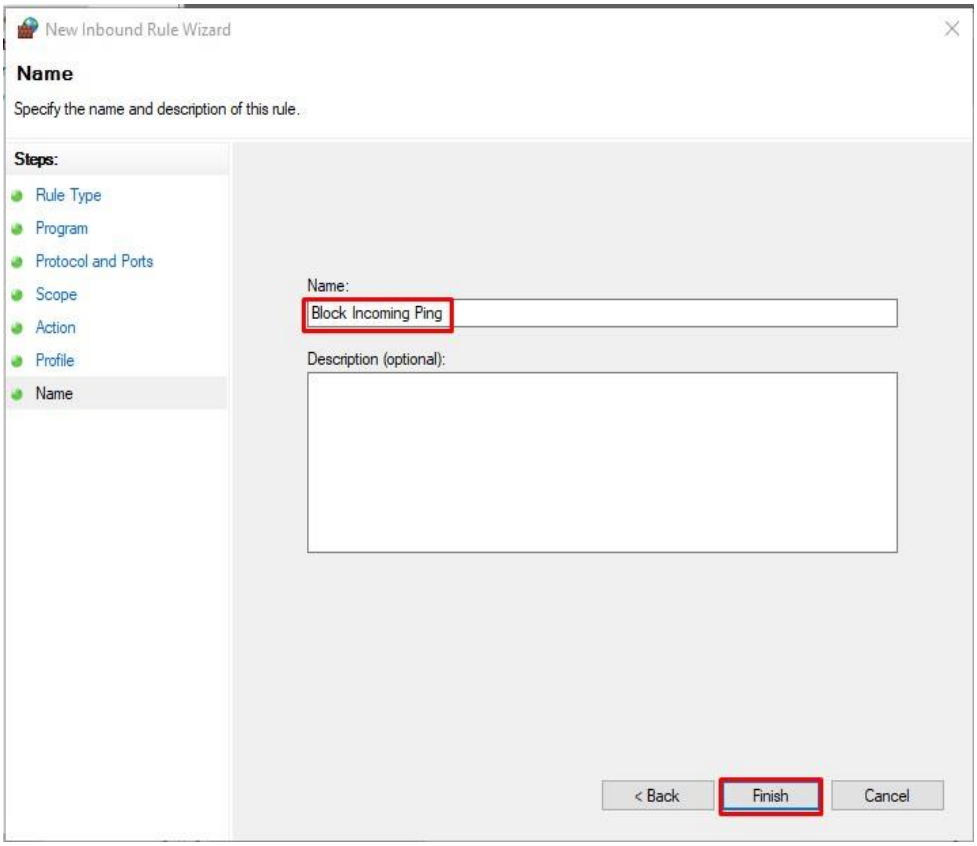
On the **Action** screen, select **"Block the connection"** and click **Next.**

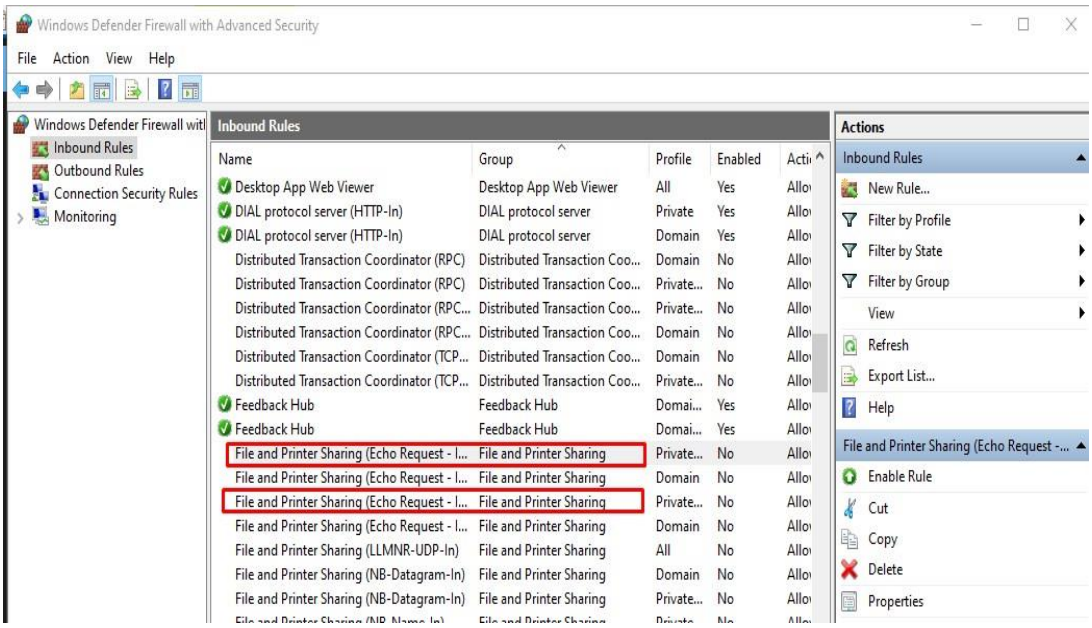


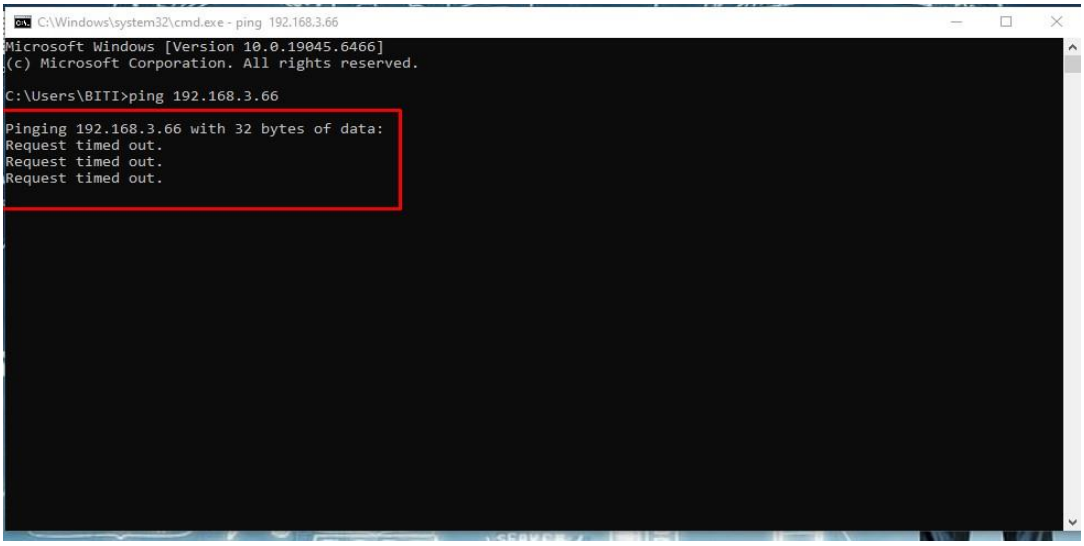
On the Profile screen, apply the rule to all profiles (Domain, Private, and Public) by checking all boxes. Click **Next**.



Finally, give the rule a descriptive name, such as **"Block Incoming Ping"**, and click **Finish.**


New Inbound Rule Wizard
Name
Specify the name and description of this rule.
Steps:
Rule Type
Program
Protocol and Ports
Scope
Action
Profile
Name
Name:
Block Incoming Ping
Description (optional):
< Back
Finish
Cancel

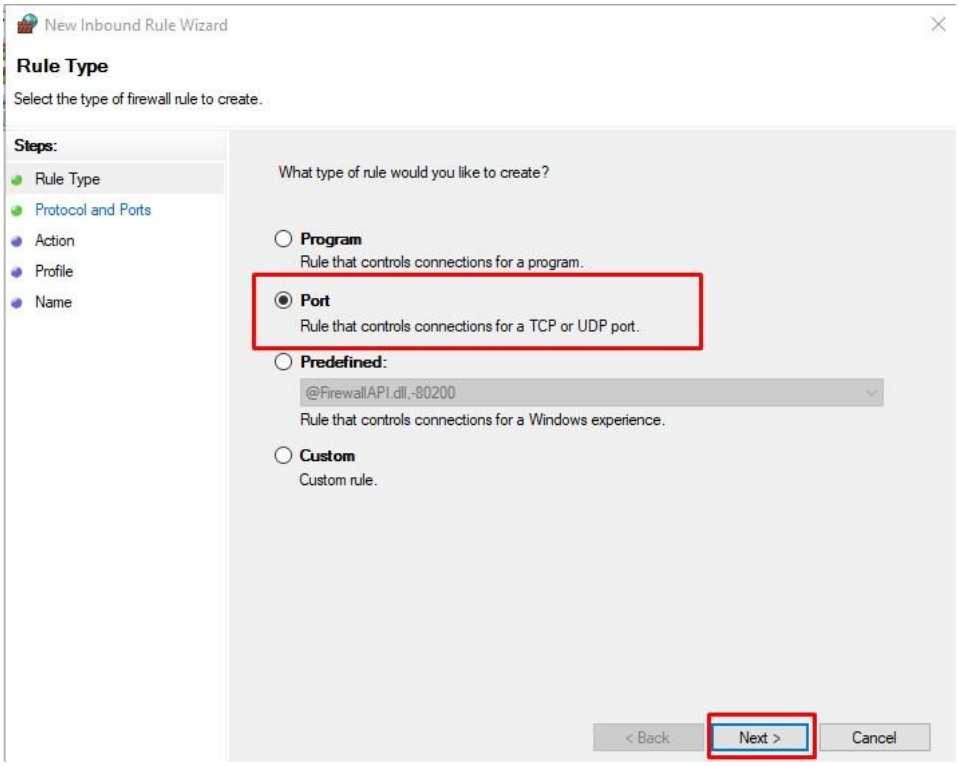

Windows Defender Firewall with Advanced Security
File Action View Help
Windows Defender Firewall with
Inbound Rules
Outbound Rules
Connection Security Rules
Monitoring
Inbound Rules
Name
Group
Profile
Enabled
Desktop App Web Viewer
DIAL protocol server (HTTP-In)
Distributed Transaction Coordinator (RPC)
Feedback Hub
File and Printer Sharing (Echo Request - I...
File and Printer Sharing (LLMNR-UDP-In)
File and Printer Sharing (NB-Datagram-In)
Actions
Inbound Rules
New Rule...
Filter by Profile
Filter by State
Filter by Group
View
Refresh
Export List...
Help
File and Printer Sharing (Echo Request -...
Enable Rule
Cut
Copy
Delete
Properties



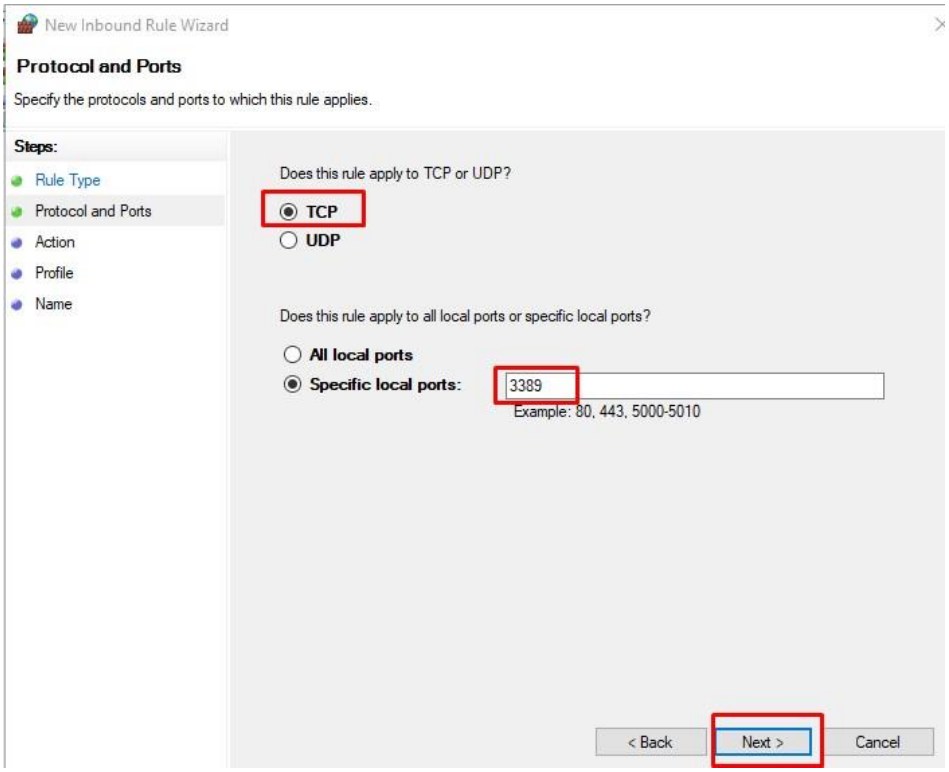
Finally Block Incoming Ping.

Blocking RDP (Remote Desktop Protocol)

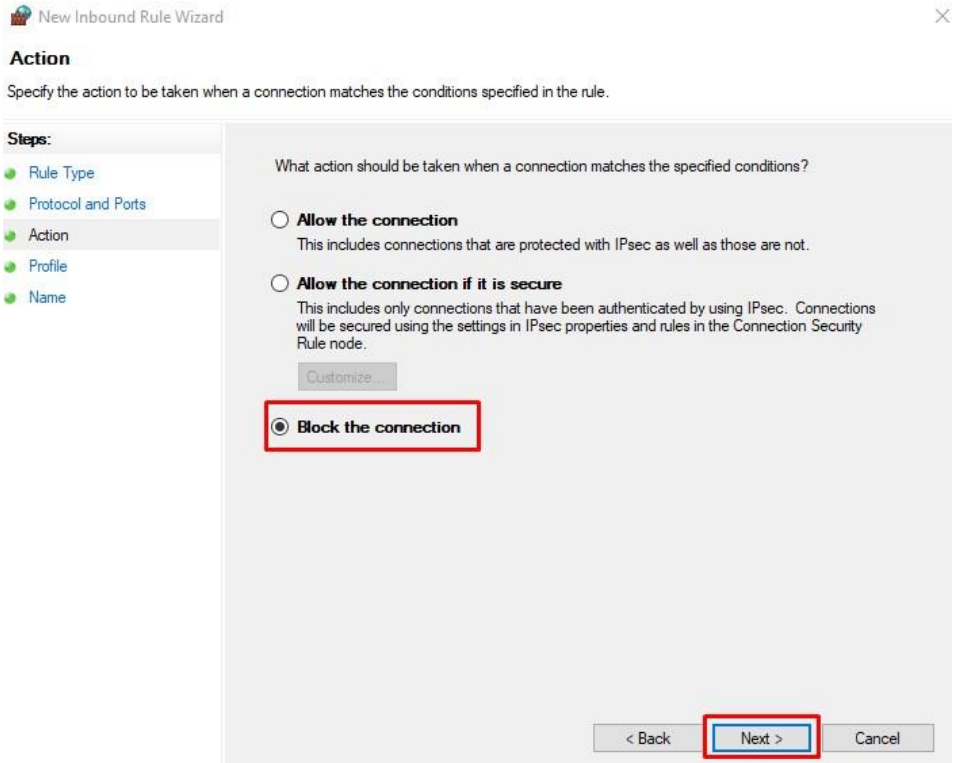
Click on "**New Rule**...". Select "**Port**" as the Rule Type.



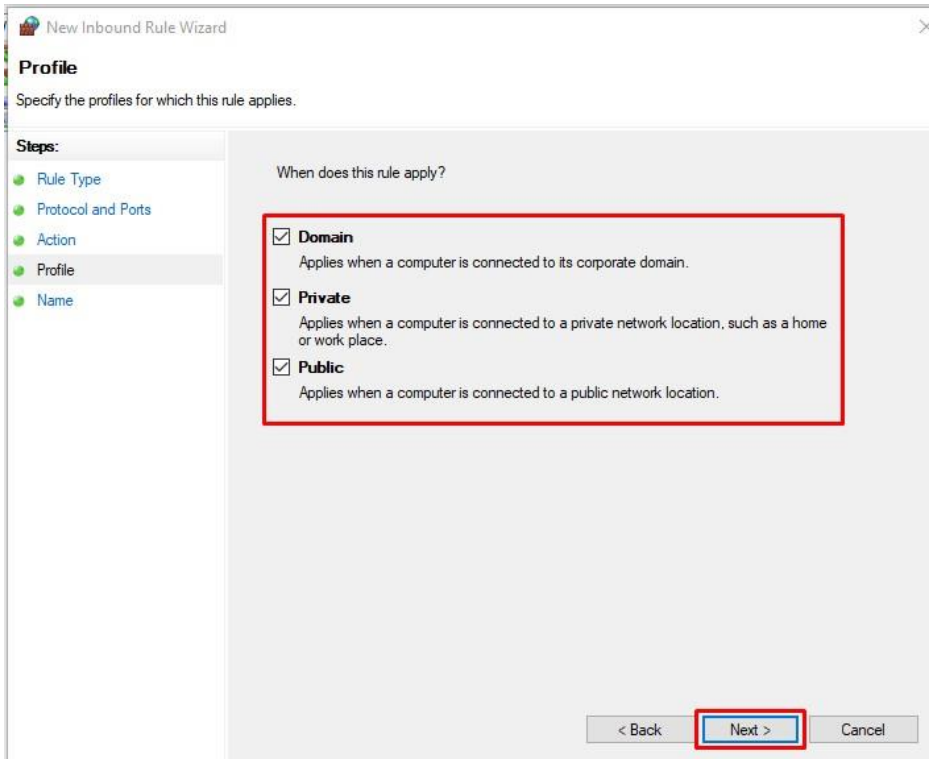
Enter **TCP** as the Protocol and **3389** (the default port for RDP) in the Specific local ports box. and **Next**.

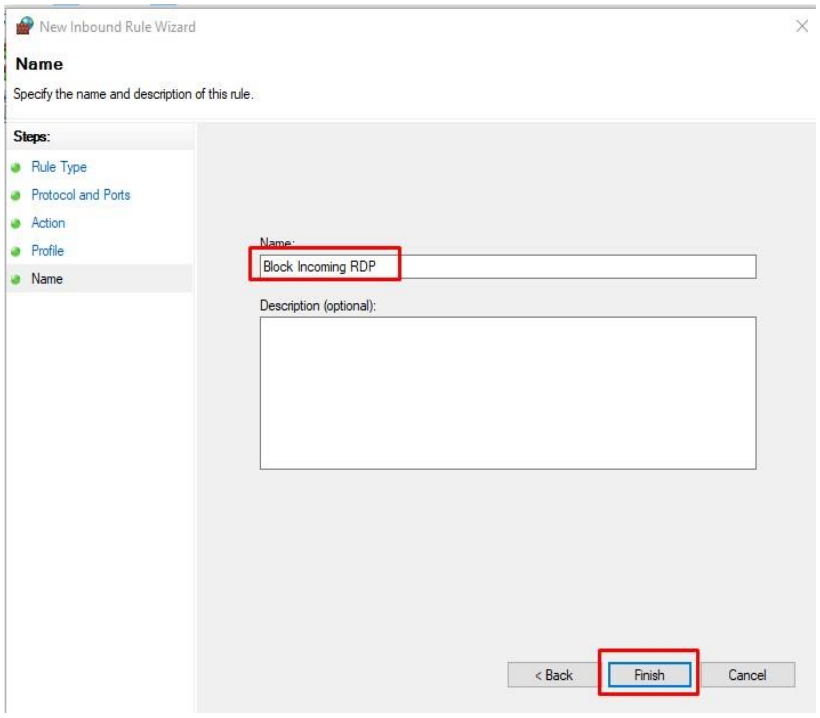


On the Action screen, select **"Block the connection".** And **Next.**

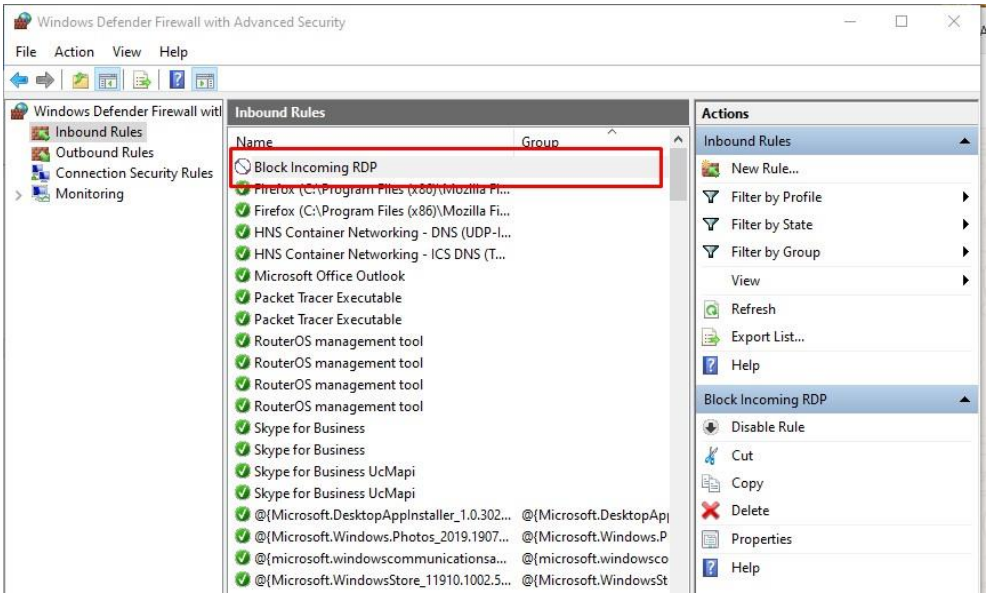


On the Profile screen, apply the rule to all profiles (Domain, Private, and Public) by checking all boxes. Click **Next**. On the Profile screen, select the profile as required, name the rule (e.g. **Block Incoming RDP**) and **Finish**.



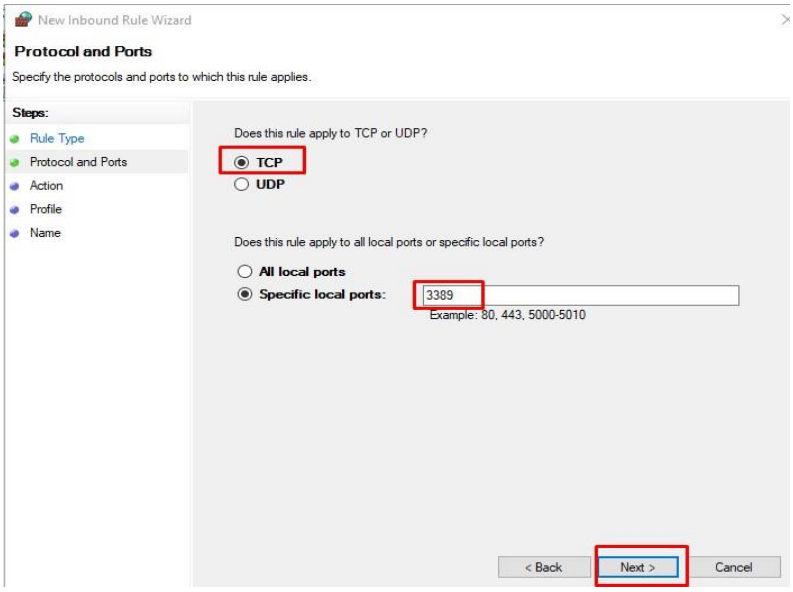


Finally **Incoming RDP Block.**



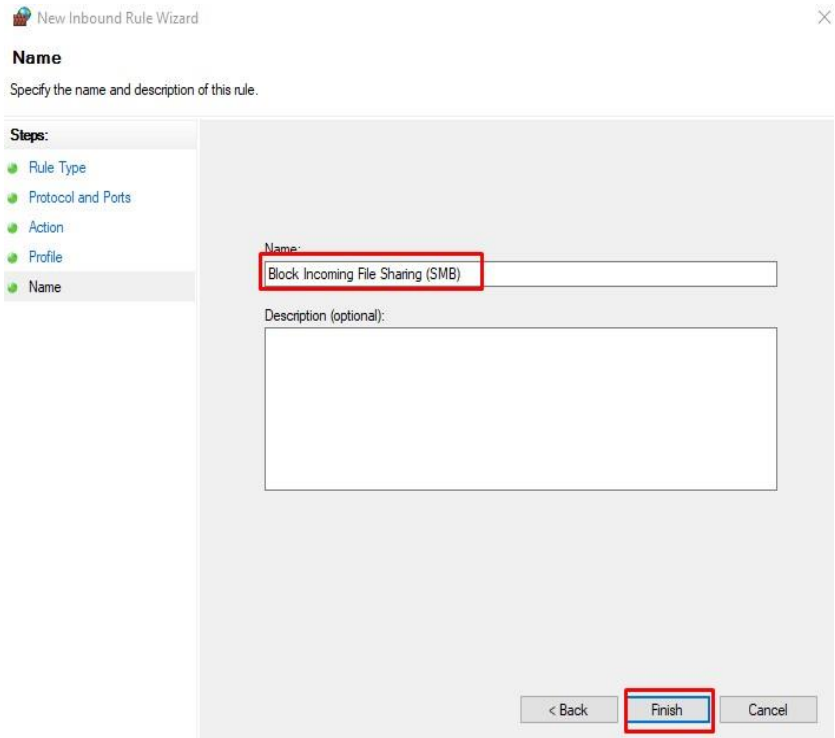
Blocking File Sharing

To block file sharing, you need to block commonly used ports (SMB). Go to **Inbound Rules** in Windows Defender Firewall with Advanced Security. In the right panel, select **New Rule**.
As with the previous RDP Block, select **Port** in Type and click **Next**.
In Protocol and Ports: Select **TCP**. Select Specific local ports and enter: **139, 445**. Click **Next**.

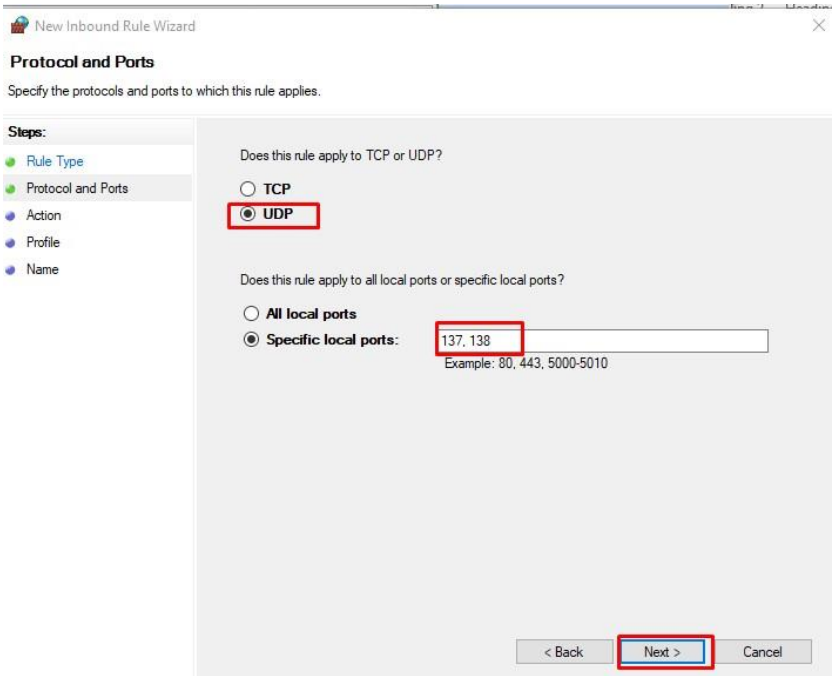


Again, according to the same rules, select **Block the connection** in (Action) and click **Next**.

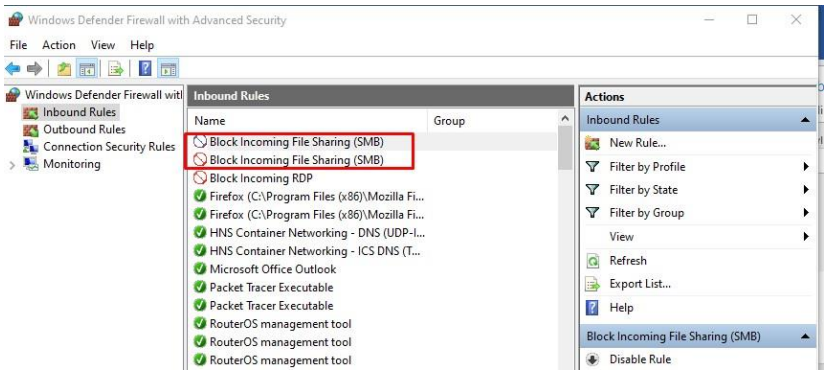
In Profile, check all **Domain, Private**, and **Public** and click **Next.** Give a Name (for example: **Block Incoming File Sharing (SMB)**) and click **Finish**.



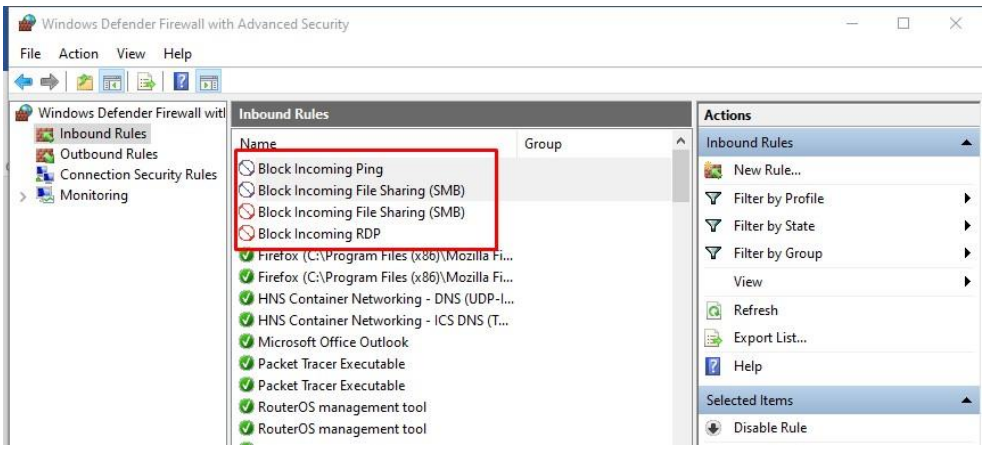
To block **UDP** ports **137 and 138**, you need to create another rule in the same process, but select UDP in the Protocol type and enter 137, 138 in the Port field.



Finally **Block Incoming File Sharing (SMB).**



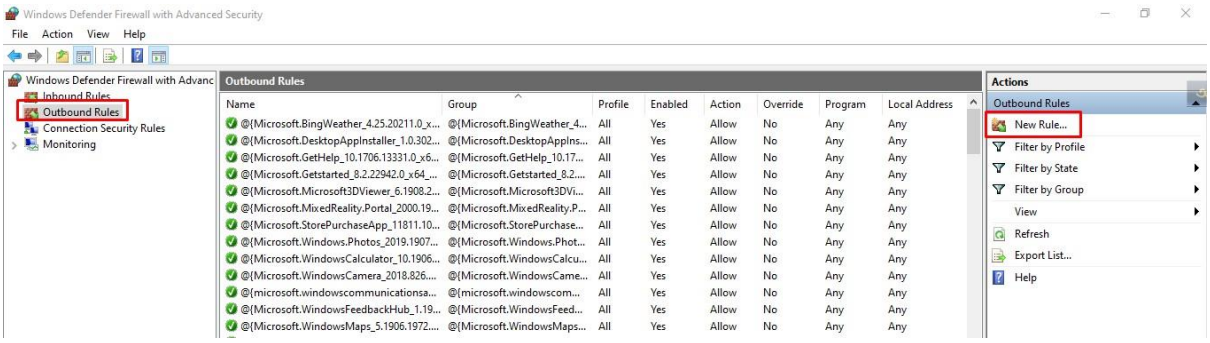
Finally Complete Task 2.



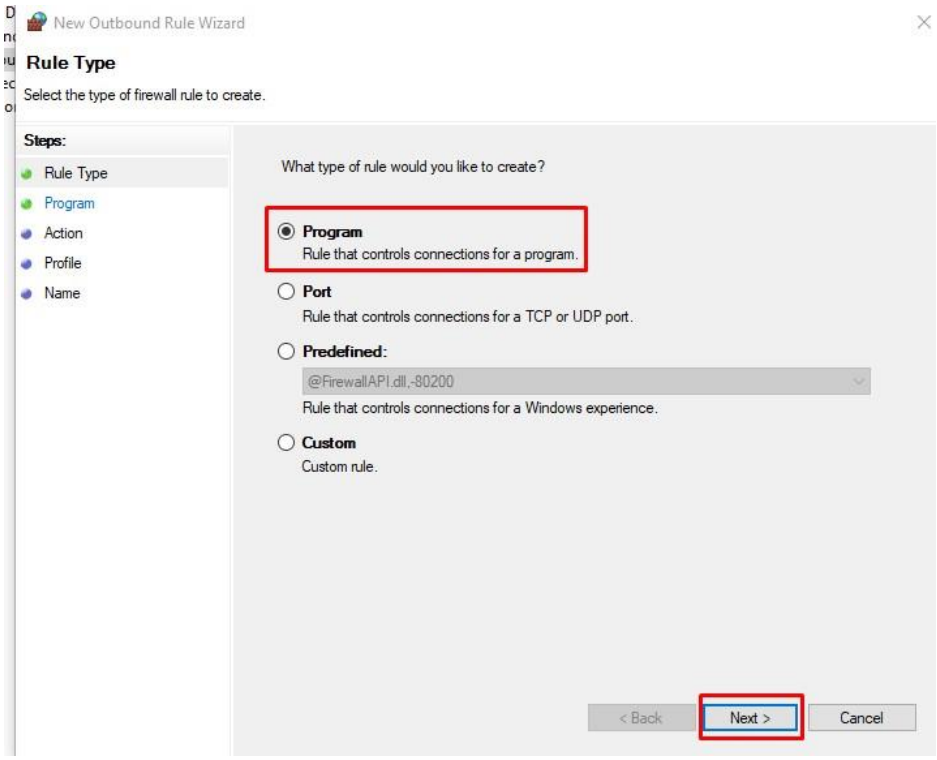
Task 3: Application Control for Google Chrome using Firewall

This will block Chrome from accessing the internet, while leaving other browsers like Microsoft Edge unaffected.

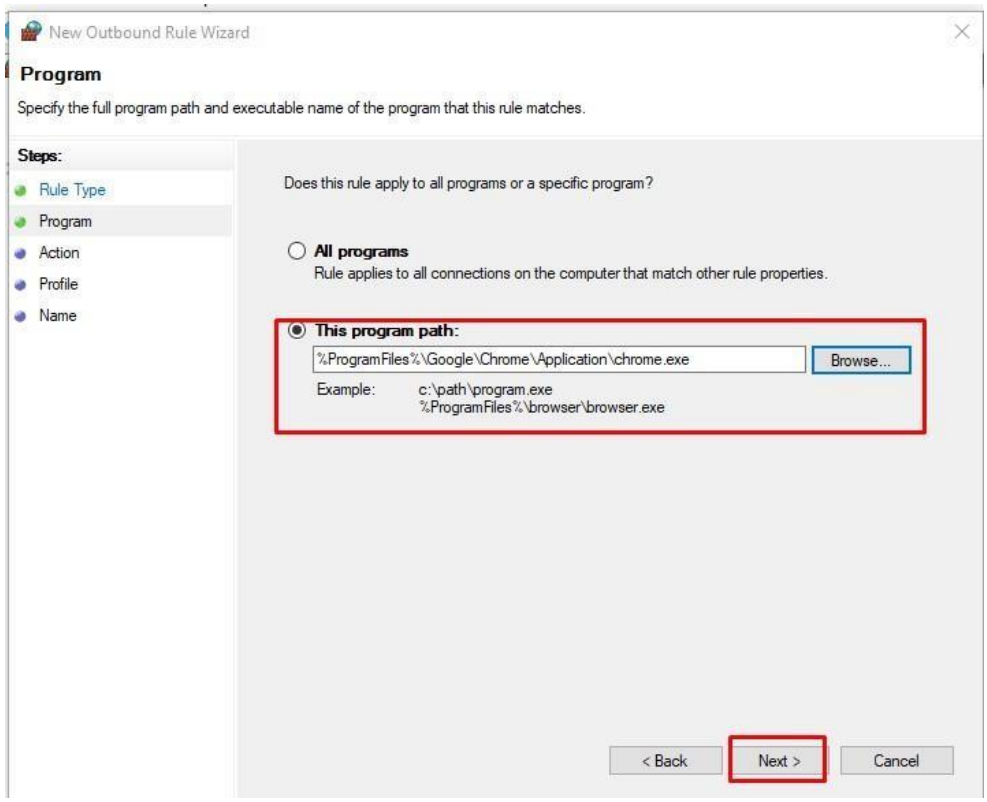
Open Windows Defender Firewall with Advanced Security (**wf.msc**). For create an Outbound Rules to Block Chrome to *Right-click* **Outbound Rules** and Select **New Rules:-**



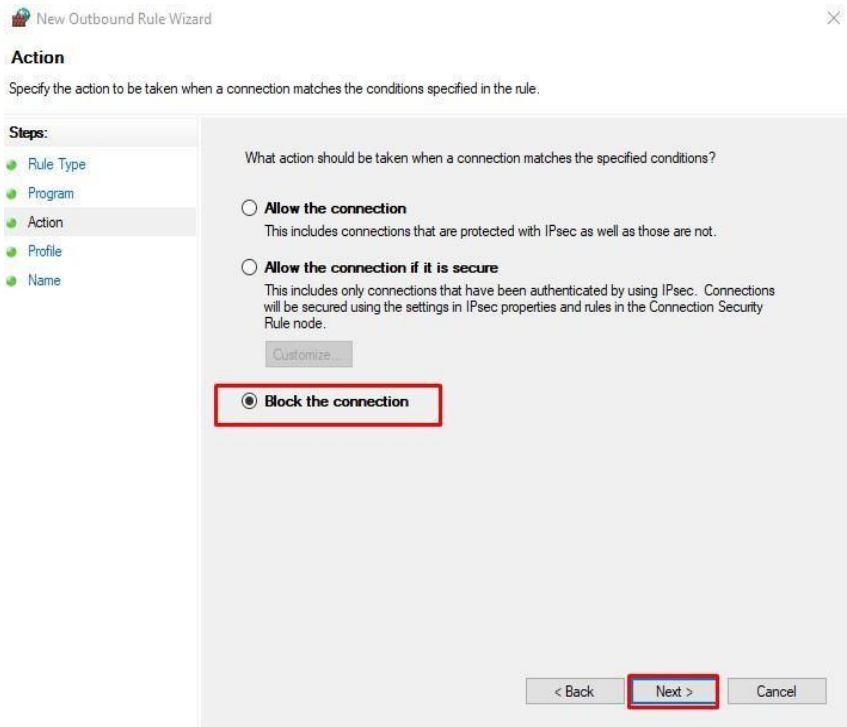
Select **Programe** and click **Next**.



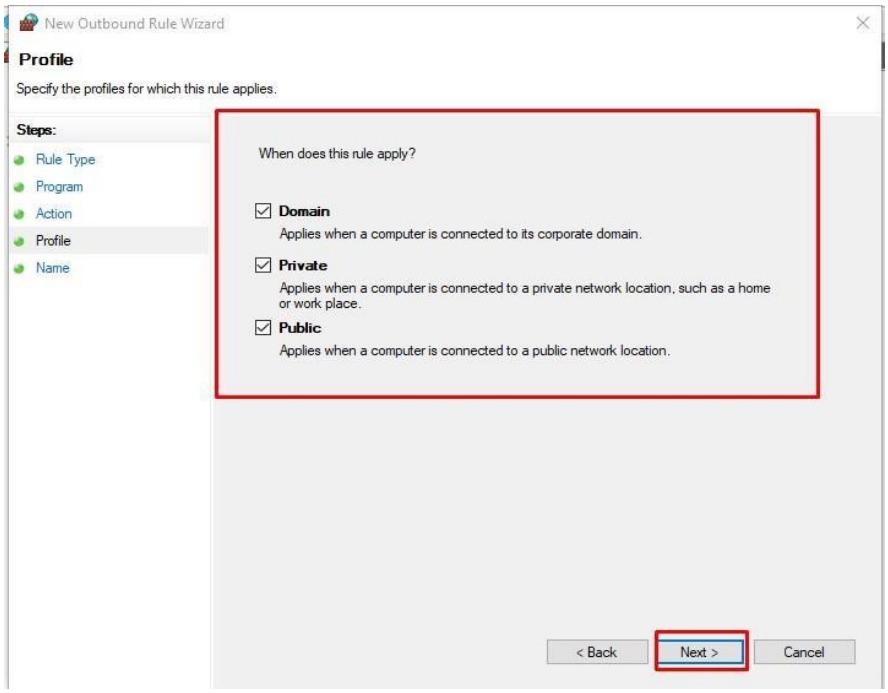
Click **Browse** and navigate to Chrome's executable (typically C:\Program Files\Google\Chrome\Application\chrome.exe). Click **Next**.



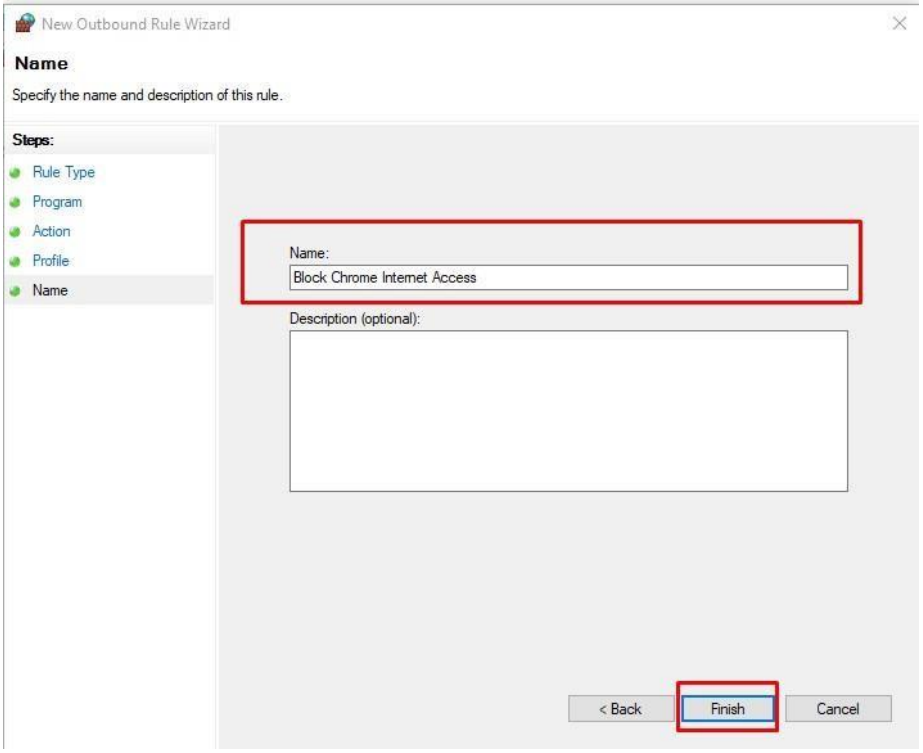
Select **Block the connection** and **Next**.



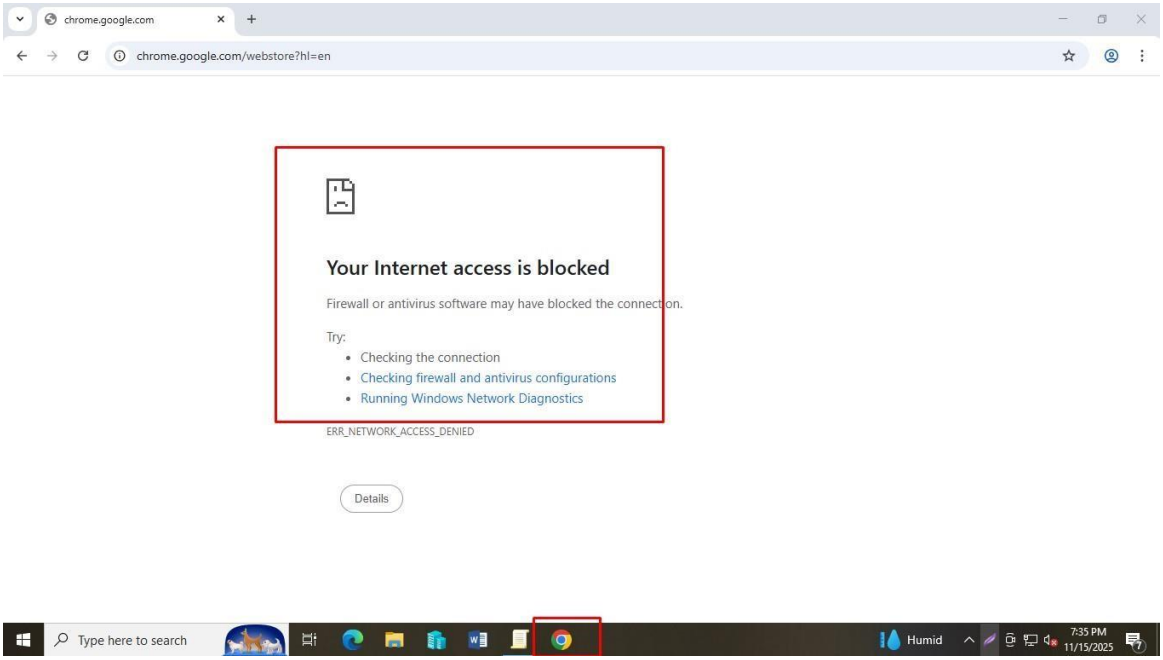
Apply to **all Profile**. click **Next**.



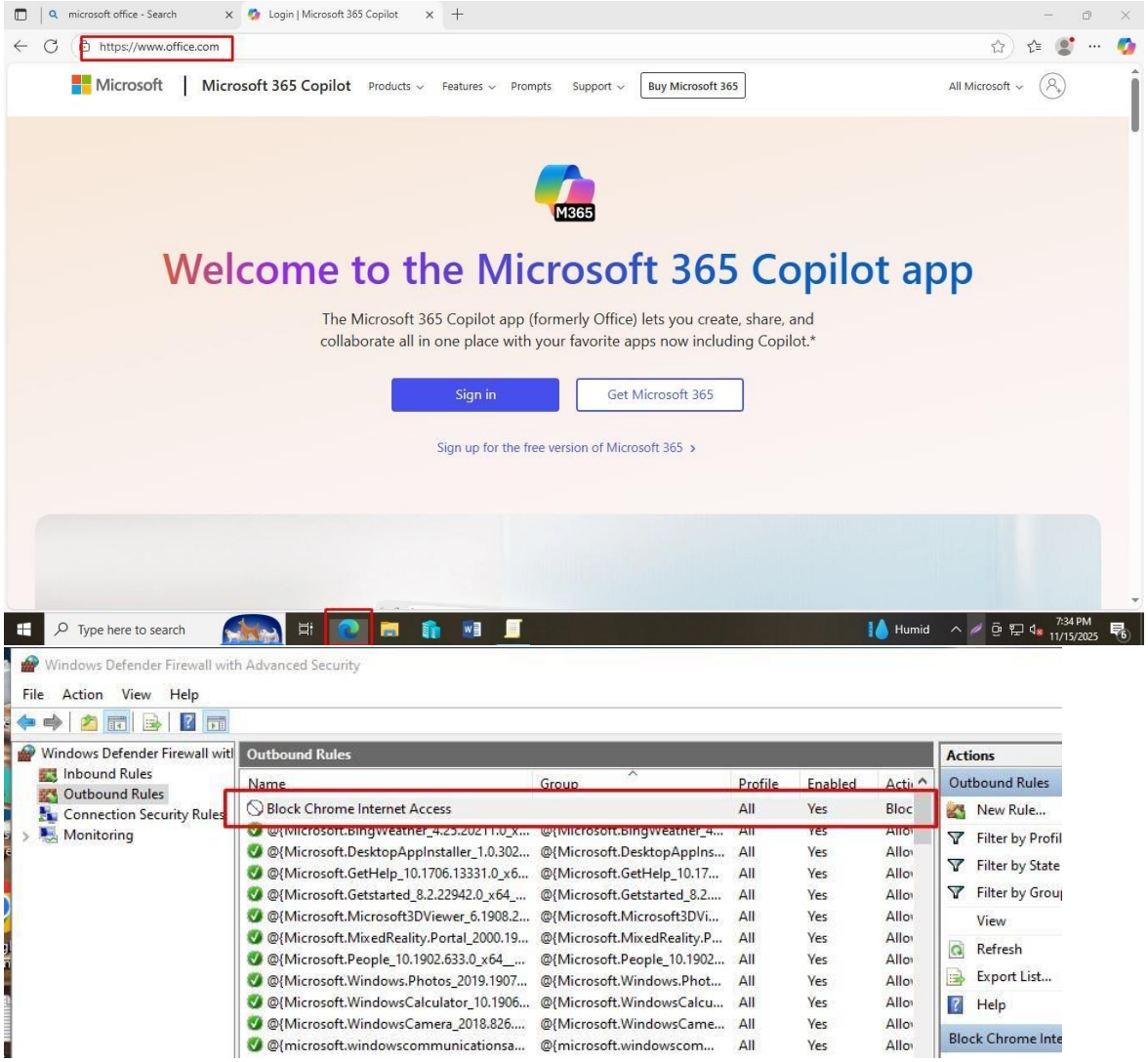
Now Enter **"Block Chrome Internet access"**. Click **finish**.



Result: Google Chrome will now fail to connect to any website, while Microsoft Edge will work perfectly.



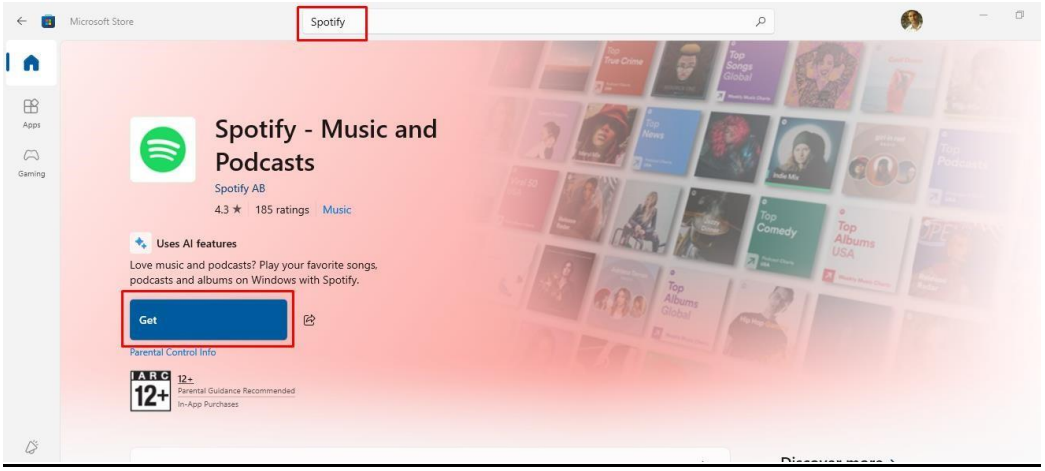
Microsoft Edge has access.



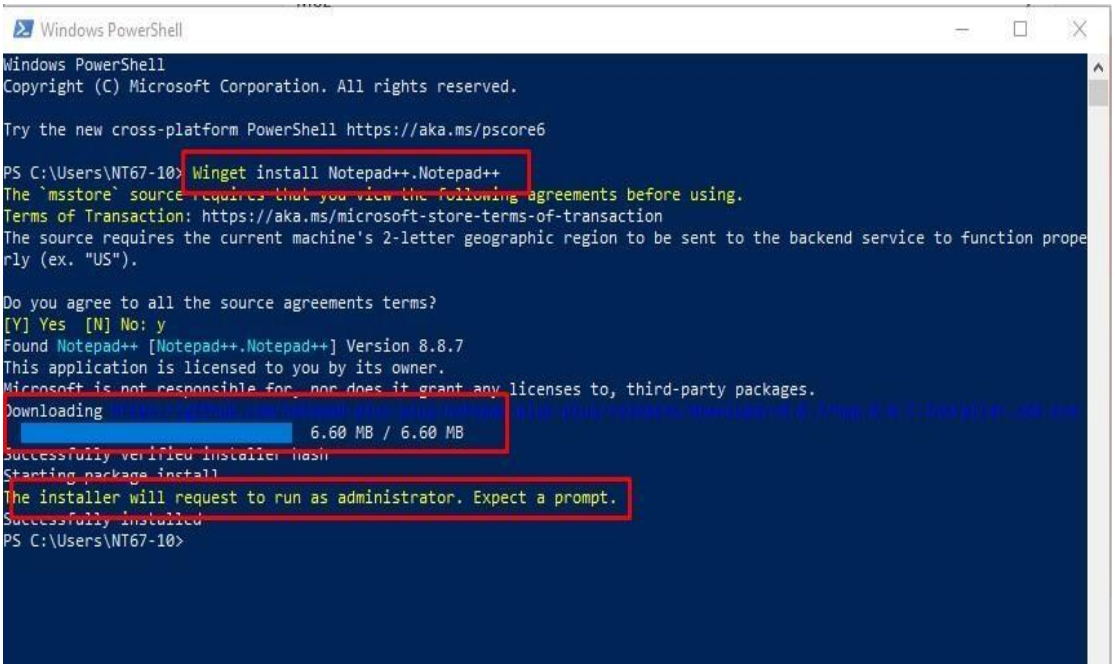
Task 4

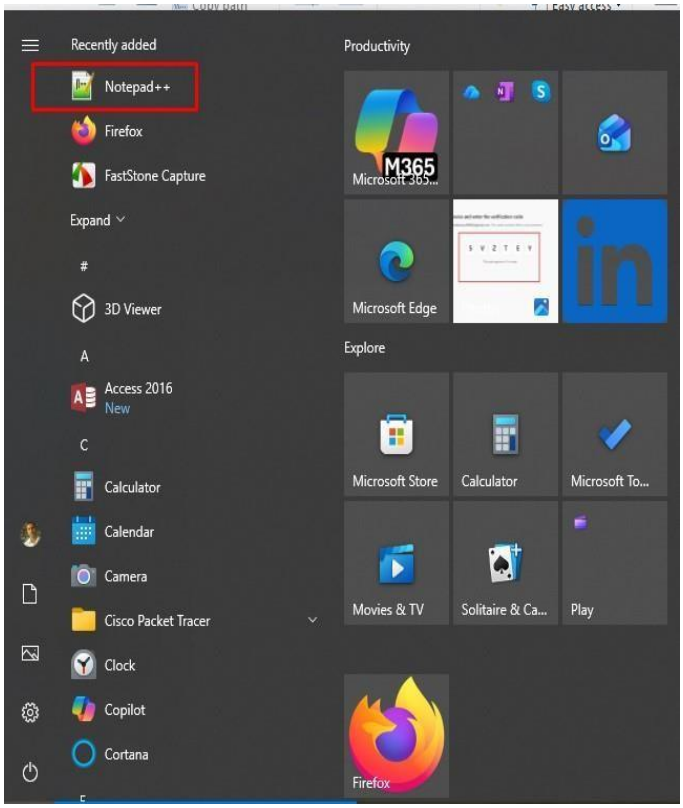
Install a Single Application from the Microsoft

Open the **Microsoft Store** application..Search for a free application, for example **"Spotify"**.Click on the application listing.Click the **"Get"** button to start the download and installation process.



Open **Command Prompt** or **PowerShell**. **And** Search **for the exact package ID for** Notepad++ : install Notepad++ using winget-





Task 5: Desktop Customization and Orientation (Virtual Desktops)

Create Virtual Desktops and Press the keyboard shortcut **Win + Tab** to open Task View.In the down-center , click “**+**” New desktop".Click the names "**Desktop 1**" and "**Desktop 2**" to rename them to "**WORK**" and "**PERSONAL**" respectively.